

Limits – 4

Upper and lower limits, subsequences, Stolz

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Theoretical minimum.

Def. $\limsup x_n = \lim_{n \rightarrow \infty} \sup_{k \geq n} x_k$ and $\liminf x_n = \lim_{n \rightarrow \infty} \inf_{k \geq n} x_k$; both always exist in $\overline{\mathbb{R}}$. A number c is a *partial limit* if some subsequence converges to c .

Theorem 1 (basic properties). $x_n \rightarrow L$ if and only if $\limsup x_n = \liminf x_n = L$; the set of partial limits is closed, and $\limsup$ and $\liminf$ are its largest and smallest elements. Moreover $\limsup(x_n + y_n) \leq \limsup x_n + \limsup y_n$, and this inequality may be strict.

Theorem 2 (Cauchy criterion and Bolzano–Weierstrass). A sequence converges if and only if it is a Cauchy sequence; every bounded sequence has a convergent subsequence, and every sequence has a monotone one.

Theorem 3 (Stolz–Cesàro and Toeplitz’s lemma). If b_n is strictly increasing and unbounded and $\frac{a_{n+1}-a_n}{b_{n+1}-b_n} \rightarrow L$, then $\frac{a_n}{b_n} \rightarrow L$. More generally, if $t_{n,k} \geq 0$, $\sum_k t_{n,k} = 1$ and $t_{n,k} \rightarrow 0$ for each fixed k , then $\sum_k t_{n,k} x_k \rightarrow \lim x_n$ (Toeplitz’s lemma); the arithmetic mean is a special case.

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1. Prove Theorem 1 in full and construct a sequence whose set of partial limits is exactly $[0, 1]$; then construct one whose set of partial limits is $\{0\} \cup \{1/k : k \geq 1\}$.
 2. Prove the Stolz–Cesàro theorem and Toeplitz’s lemma, deduce from the lemma the theorem on arithmetic means, and give an example showing that the converse of Stolz fails.
 3. Prove that if $x_{n+1} - x_n \rightarrow 0$, then the set of partial limits of (x_n) is an interval (possibly infinite or degenerate). Give an example of a bounded sequence with $x_{n+1} - x_n \rightarrow 0$ which has no limit.

4. (Mexmat, 2016) For $S \subseteq \mathbb{N}$ define the upper density $D^*(S) = \limsup_{n \rightarrow \infty} \frac{|S \cap \{1, \dots, n\}|}{n}$, and suppose $D^*(S) > 0$. Is it always possible to find $n \in \mathbb{N}$ for which $(S - n) \cap S$ has positive upper density?
5. (CIIM 2014) Let $x_n \in [0, 1]$ for all n .
 - a) Prove that there is $C > 0$ such that for every positive integer r one can find $m \geq 1$ and $n > m + r$ with $(n - m)|x_n - x_m| \leq C$.
 - b) Prove that for every $C > 0$ there are a sequence $x_n \in [0, 1]$ and an integer r such that $(n - m)|x_n - x_m| > C$ whenever $m \geq 1$ and $n > m + r$.